

MERN Stack Technical Assignment - Inventory & Barcode Management

Dear Candidate,

Thank you for participating in our interview process.

As the next step, please complete the following MERN Stack technical assignment within 2-3 days.

Assignment: Inventory CRUD and Barcode Management

Build a small web application with the following features:

1. Inventory Management

- Create, view, update and delete inventory items.
- Include: Item Name, SKU, Category, Price, Quantity and Storage Location.
- SKU must be unique.
- Price and quantity cannot be negative.
- Provide search by item name or SKU.
- Add pagination to the inventory list.
- Request confirmation before deleting an item.

2. Barcode Functionality

- Automatically generate a Code 128 barcode using the item's SKU.
- Display the barcode on the item details page.
- Allow users to download or print the barcode.
- Provide an input field for barcode scanning or manual entry.
- Display the corresponding item when a barcode is entered.
- Show an appropriate error if the barcode is invalid or not found.

Technical requirements:

- MongoDB with Mongoose
- Express.js and Node.js
- React.js
- REST APIs
- Frontend and backend validation
- Responsive user interface
- Proper error handling and HTTP status codes

Expected deliverables:

- Git repository link
- Complete frontend and backend source code
- README with setup instructions
- .env.example file

- Sample inventory data

Please submit the completed assignment within deadline.

The assignment will be evaluated based on functionality, code quality, database and API design, validation, error handling, UI responsiveness and documentation.

Please confirm receipt of this assignment. Feel free to contact us if you have any questions regarding the requirements.

Best regards,
Tech Team
Atmik Bharat Industries Pvt. Ltd.